

of 32MHz to 80MHz with math acceleration and multiple configurations of integrated analogue signal-chain components, including the industry's first zero-drift operational amplifier on an MCU and precision 12-bit, 4MSPS analogue-to-digital converters. The MCUs can help save design time with software, design support resources and coding tools, which also include graphical tools that streamline device configuration—all created to help designers code once and then scale across future MSPM0 based designs.

Texas Instruments
<https://www.ti.com>

BOARDS AND MODULES

Rugged DC-DC converters

PQQC3-OS from CUI Devices is a series of regulated DC-DC converters for sensitive applications. The converters are compliant with 62368 safety standards for CE and UKCA. Both the power



converters feature a 2:1 input range along with features such as remote on/off control, short-circuit and over-current protection. The PQQC3-OS series can provide power up to 3W and can operate over a wide range of temperatures from -40 to +85°C. These converters are available in an industry standard 8-pin SIP package measuring just 22mm × 8.2mm × 12.8mm. They are suitable for use in a variety of industrial, test & measurement, and telecommunication applications.

CUI Devices
<https://www.cui.com>

Two-stage integrated power

BMR510 from Flex Power is a 2-phase integrated power stage that delivers a continuous total output current of 40A per phase, or 80A total, with an efficiency of up to 90%. Compared to the previous model, this upgraded module offers higher inductance in a smaller package, allowing a lower switching frequency operation. The module al-

lows for remote monitoring and comes with protective features, including output over-current and over-temperature protection. The BMR510 is optimised for top-side cooling and is halogen-free. It can accept tri-state pulse width modulation inputs from the chosen controller and a separate enable input is provided. The device is suitable for providing power for silicon based chips that have a high current requirement, such as GPUs, CPUs, IPUs, high-end FPGAs, and high-performing ASICs often found in the latest AI applications.

Flex Power Module
<http://www.flexpowermodules.com>

ELECTRONICS MANUFACTURING SOLUTIONS

Rework system for PCBs

VJ Electronix's LT120 is a rework system that can handle PCBs for servers, backplanes, and extra-large boards. The rework process typically involves several steps, which may include inspection, preparation, removal,



replacement, and testing. The large work area and high-resolution imaging in the LT120 system enable it to work with very large BGA components on large PCBs. It allows fast and accurate alignment of BGAs and other packages up to 120 mm × 120mm, which enables a higher system throughput. LT120 employs an ultra-high-resolution vision system, which provides the accuracy needed to deliver reliable rework for these very high-value components, making it suitable for extra-large servers, backplanes and boards, and/or very large component applications. It utilises a new vision system with an increased field of view for aligning these very large BGAs. The intuitive alignment GUI provided by the company allows the user to quickly identify the component, large or small, then zoom in to the corners for a simplified, yet

accurate alignment, thus increasing system throughput.

VJ Electronix
<https://www.vjelectronix.com>

T&M EQUIPMENT

Compact storage oscilloscope

SDS6000L is a low-profile digital storage oscilloscope series from SIGLENT. The digital oscilloscopes feature up to 8 analogue channels and 16 digital



channels in compact packaging. The series is available in bandwidths of 2GHz, 1GHz, and 500MHz. The oscilloscopes employ an innovative digital trigger system with high sensitivity and low jitter. The trigger system supports multiple powerful triggering modes, including serial bus triggering. SDS6000L series can be used as stand-alone oscilloscopes by connecting to an external display and a mouse.

SIGLENT
<https://int.siglent.com>

EMI test receiver

R&S EPL1000 from Rohde & Schwarz is an EMI test receiver for emission measurements up to 30MHz. It is CISPR 16-1-1 compliant and suitable



for certification measurements. The instrument reduces uncertainty in pre-compliance measurement tasks. Moreover, the fast time domain scan lets EPL1000 check all frequencies in CISPR bands A or B in a single shot for quick and seamless measurements over extended periods of time, when required. The company provides a user-friendly GUI that can enable EMC engineers to quickly find infrequent emissions and gain a good overview. EPL1000 is capable of giving reproducible and standard compliant test results, ensuring high product quality.

Rohde & Schwarz
<https://www.rohde-schwarz.com>